

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 2 – CONCEPT MAPPING

|                  |                                                                                                                                                                |               |                                     |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------|
| Course Code:     | CSA06                                                                                                                                                          | Course Title: | Design and Analysis of Algorithms   |
| CO Assessed:     | CO1 – Precision (BL1, BL2, BL3)                                                                                                                                | Assessment:   | Assessment Tool 2 – Concept Mapping |
| Weightage:       | 20% of CIA                                                                                                                                                     | Total Marks:  | 100 (2 Questions × 50 Marks)        |
| CO1 Statement:   | <i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i> |               |                                     |
| Student Name:    | S. Jaxon Rizal                                                                                                                                                 | Reg. No.:     | 192511421                           |
| Section / Batch: |                                                                                                                                                                | Date:         |                                     |

### Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

| Question Type   | Focus Area                                                                                                                      | Questions | Marks            |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------|-----------|------------------|
| Concept Mapping | Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method | Q1 – Q2   | Q1 –50<br>Q2 –50 |
| GRAND TOTAL:    |                                                                                                                                 | 100 Marks |                  |

### SECTION A – CONCEPT MAPPING

#### Code of Conduct

I S. Jaxon / 192511421 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the Student: \_\_\_\_\_

*S. Faxon***RUBRICS FOR CALCULATION**

| Criteria                  | Weightage | Level 4 – Exemplary                                                         | Level 3 – Proficient                              | Level 2 – Developing                                        | Level 1 – Beginning                     |
|---------------------------|-----------|-----------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------|-----------------------------------------|
| Concept Identification    | 25%       | Identifies all key concepts from literature accurately and comprehensively  | Identifies most key concepts with minor omissions | Identifies limited concepts; some are irrelevant or missing | Fails to identify essential concepts    |
| Linkages                  | 25%       | Establishes clear, meaningful, and accurate relationships between concepts  | Shows correct relationships with minor gaps       | Relationships are weak, unclear, or partially incorrect     | No meaningful relationships established |
| Organization              | 20%       | Concepts are well-structured hierarchically with logical flow and grouping  | Mostly organized with minor structural issues     | Some organization but lacks clear hierarchy                 | No logical organization or structure    |
| Integration of Literature | 20%       | Integrates multiple studies effectively to show connections and comparisons | Shows some integration of literature              | Limited integration; mostly isolated concepts               | No integration of literature evidence   |
| Clarity                   | 10%       | Concept map is clear, neat, visually structured, and easy to interpret      | Generally clear with minor readability issues     | Clarity is reduced; difficult to interpret in parts         | Poorly presented and hard to understand |

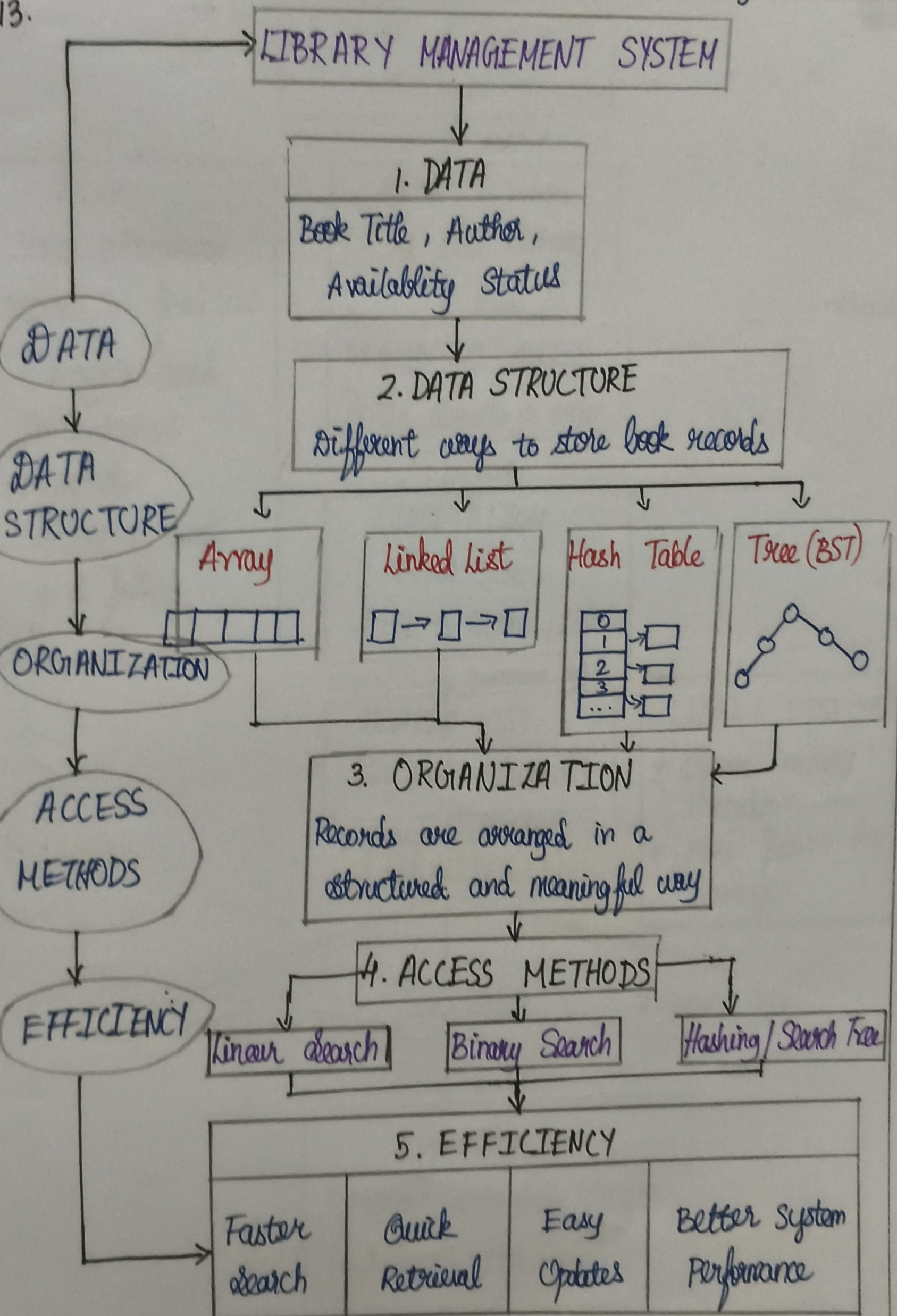
**Marks Evaluation:**

| Question No.        | Concept Identification (25%) | Linkages (25%) | Organization (20%) | Integration of Literature (20%) | Clarity (10%) | Total Marks (Each – 50) |
|---------------------|------------------------------|----------------|--------------------|---------------------------------|---------------|-------------------------|
| Q1                  |                              |                |                    |                                 |               |                         |
| Q2                  |                              |                |                    |                                 |               |                         |
| Overall Total (100) |                              |                |                    |                                 |               |                         |

| Faculty In-charge | Course Coordinator | Head of Department |
|-------------------|--------------------|--------------------|
| Signature: _____  | Signature: _____   | Signature: _____   |
| Date: _____       | Date: _____        | Date: _____        |



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# STUDENT ATTENDANCE SYSTEM

**ARRAY**  
Stores attendance records in fixed size.  
Elements stored in contiguous memory.  
Access is simple and fast.

ARRAY

(Fixed Size Storage)

SEQUENTIAL ACCESS  
(Access elements in order)

LIMITATION

Fixed size,  
Hard to insert / delete

LINKED LIST  
(Dynamic Allocation)

FLEXIBILITY

Easy Insert

Easy Delete

Dynamic Size

Application

Frequent Attendance Updates  
& Record Management

**LINKED LIST**

- \* Dynamic memory allocation
- \* More flexible than arrays.